

ANIRUDH MAKULURI

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SUMMARY

Full-stack software engineer with 2+ years of experience building production web, mobile, backend, and cloud systems. Early engineer at a startup where I helped scale a real-time collaboration platform to 12K+ users, owning APIs, deployments, performance, and user-facing features. Recently developed SmartDeploy and SD-Artifacts, LLM-assisted infrastructure tools that analyze codebases, generate build plans, and execute cloud workflows.

EDUCATION

M.S Computer Science <i>Arizona State University, Tempe, AZ (ASU)</i>	August 2024 – May 2026 GPA: 4.00/4.00
B.E. Electronics & Instrumentation <i>Birla Institute of Technology and Science Pilani (BITS Pilani)</i>	August 2018 – May 2022

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, Go, SQL, Java
Frontend/Mobile: React, React Native, Next.js, Zustand, TanStack Query, Redux, Tailwind CSS
Backend/APIs: Node.js/Express, FastAPI, REST APIs, WebSockets, Socket.IO, Microservices
Databases: PostgreSQL/Supabase, Firestore, MongoDB, DynamoDB, Neo4j
Cloud/DevOps: AWS (EC2, SSM, ECR, ALB, ECS), GCP, Docker, Terraform, GitHub Actions CI/CD, Kubernetes
AI/LLM: RAG, embeddings, vector search, LangChain, LangGraph, AWS Bedrock, TensorFlow, PyTorch

PROFESSIONAL EXPERIENCE

Shardings — Full-Stack Software Engineer (Early Engineering Team)	June 2022 – August 2023
- Scaled a real-time video conferencing platform across web and mobile to 12K+ users, improving production reliability using React, React Native, Node.js, Firebase, and GCP - Redesigned real-time session synchronization for stateless servers, preventing context loss for late-joining users by replaying chat messages, polls, files, and shared meeting context from Redis-backed state. - Designed 15 reusable React + Tailwind components across dashboard and meeting workflows, helping migrate the web app from legacy vanilla JavaScript to a responsive design system aligned with Figma specs. - Integrated Google Cloud Speech-to-Text and ChatGPT API to transcribe and summarize live standups, eliminating 5 hours/week of manual note-taking per team and making meeting recaps searchable. - Owned Node.js backend services powering 35+ REST APIs and 30+ WebSocket streams; resolved critical WebSocket and media server incidents with 1 hour MTTR while maintaining production reliability.	
Shardings — Software Engineer Intern	July 2021 – December 2021
- Streamlined full payment gateway with Razorpay and Stripe integrations supporting 3 pricing tiers (Basic, Standard, Pro); designed secure webhook handling, idempotency, and error recovery, handling 100+ transactions per month with 99% success rate. - Introduced Google Analytics to track 8+ user interaction and error events (room creation, joins, failures), improving observability into frontend behavior and enabling faster debugging of user-facing issues.	

PROJECTS

SmartDeploy — Preview-Driven Deployment Platform	July 2025 – Present
- Introduced a deployment preview system(Next.js, Node.js) that helps users understand how their app will go live before deployment, including build steps, hosting target, routing, DNS, secrets, monitoring, and rollback. - Automated AWS deployment setup by analyzing each repository and choosing the right hosting path: ECS Fargate + ALB for backend apps and S3 + CloudFront for static websites. - Built a production-style release pipeline using CodeBuild, ECR, ECS Fargate, ALB, S3, CloudFront, Route 53, Secrets Manager, and CloudWatch, handling builds, DNS, env vars, health checks, monitoring, and rollback across 105+ tracked runs.	
SD-Artifacts — Infrastructure Generation Engine	January 2026 – March 2026
- Developed a LangGraph + AWS Bedrock pipeline (FastAPI) that analyzes GitHub repositories, generates Railpack-verified build plans, and performs autonomous build repair with up to 3 validation retries. - Cut repeat-analysis overhead by 90% using commit-scoped Supabase caching; shipped a read-only MCP server exposing analysis history, generated artifacts, and repair logs to AI agents.	
Hoplio — Secure Realtime Communication Platform	November 2023 – January 2026
- Engineered a cross-platform realtime communication platform with React Native, React.js, Node.js, Socket.IO, and Redis, achieving sub-500ms interactions across mobile and web while refactoring the backend for stateless horizontal scaling. - Architected a secure, offline-first synchronization layer with end-to-end encryption, retries, and conflict resolution, ensuring ordered, reliable consistency under unstable networks and supporting an AI assistant with semantic retrieval.	

ACHIEVEMENTS & CERTIFICATIONS

- 1st Place (Auth0 AI Agents) & 2nd Place (Google Track)** — Innovation Hacks 2.0, ASU (600+ participants), April 2026
Developed *TravelMate*, AI travel companion with real-time voice call interface for itinerary assistance
- Certifications:** Oracle Generative AI Professional; ML Specialization (Stanford/DeepLearning.AI); Oracle Certified Associate